Enforcing Integrity Constraints

What is Enforcing Integrity Constraints?

- Enforcing integrity constraints means applying rules to maintain correct and valid data in a database.
- The Database Management Systems automatically checks these rules during:
  - Insert
  - Update
  - Delete operations

Definition

Enforcing Integrity Constraints:
The process of ensuring that database rules are followed to maintain data accuracy and consistency.

Why Integrity Constraints Are Enforced?

1. To avoid invalid data
2. To maintain consistency
3. To prevent duplication
4. To protect relationships between tables
5. To improve database reliability

Main Integrity Constraints Enforced in DBMS

1. Domain Constraint
2. Key Constraint
3. Entity Integrity Constraint
4. Referential Integrity Constraint

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1. Enforcing Domain Constraint

Definition

- Ensures attribute values belong to valid domain.

Example

- Age must be between 1 and 100.

How DBMS Enforces It

6. Checks data type
7. Checks value range
8. Rejects invalid values

Example

Age

25 ✓

-5 ❌

- Negative age is rejected.

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2. Enforcing Key Constraint

Definition

- Ensures key values are unique.

Example

- Student_ID must not repeat.

How DBMS Enforces It

9. Checks duplicate values
10. Rejects repeated keys
11. Maintains uniqueness

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Example

Student_ID

101 ✓

101 ❌

- Duplicate key is not allowed.

3. Enforcing Entity Integrity Constraint

Definition

- Primary key cannot contain NULL values.

How DBMS Enforces It

12. Prevents NULL primary key
13. Ensures each row has identity
14. Rejects incomplete records

Example

Student_ID

NULL ❌

- NULL primary key is rejected.

4. Enforcing Referential Integrity Constraint

Definition

- Foreign key must match existing primary key.

How DBMS Enforces It

15. Checks referenced table
16. Prevents invalid references
17. Maintains table relationships

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Example

Department Table

Dept_ID
D1

Student Table

Student_ID	**Dept_ID**
101	D1 ✓
102	D5 ❌

- D5 does not exist in Department table.

Methods Used to Enforce Constraints

1. Using Primary Key

18. Ensures uniqueness
19. Prevents NULL values

2. Using Foreign Key

20. Maintains table relationships
21. Prevents invalid references

3. Using CHECK Constraint

22. Verifies condition before insertion

Example

CHECK (Age >= 18)

4. Using NOT NULL Constraint

23. Prevents empty values

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Example

Name VARCHAR(20) NOT NULL

5. Using UNIQUE Constraint

24. Prevents duplicate values

Example

Email VARCHAR(50) UNIQUE

6. Using DEFAULT Constraint

25. Assigns default value automatically

Example

Country DEFAULT 'India'

Actions Taken During Constraint Violations

Insert Operation

26. Invalid data insertion rejected

Update Operation

27. Invalid updates blocked

Delete Operation

28. Prevents deletion if related records exist

Referential Integrity Actions

1. CASCADE

29. Automatically updates/deletes related records

2. SET NULL

30. Sets foreign key to NULL

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3. RESTRICT

31. Prevents deletion of referenced row

4. NO ACTION

32. Rejects operation causing violation

Advantages of Enforcing Integrity Constraints

33. Maintains data accuracy
34. Prevents inconsistent data
35. Improves database quality
36. Reduces redundancy
37. Supports reliable transactions
38. Protects relationships
39. Improves security
40. Simplifies maintenance

Additional Important Points

41. Constraints are checked automatically by DBMS.
42. Integrity enforcement improves consistency.
43. Primary key enforcement avoids duplicates.
44. Foreign key enforcement maintains relationships.
45. CHECK constraints validate conditions.
46. NOT NULL prevents empty data.
47. UNIQUE ensures distinct values.
48. Constraint violations generate errors.
49. Integrity constraints are essential for relational databases.
50. Proper enforcement improves database performance.

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Real-Time Example

Banking Database

Account Table

- Account_Number → Primary Key

Transaction Table

- Account_Number → Foreign Key
- Transactions allowed only for valid accounts.

Short Definitions for Exams

Enforcing Integrity Constraints

Applying rules to ensure valid and consistent data in the database.

Referential Integrity Enforcement

Ensuring foreign key values match primary key values.

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